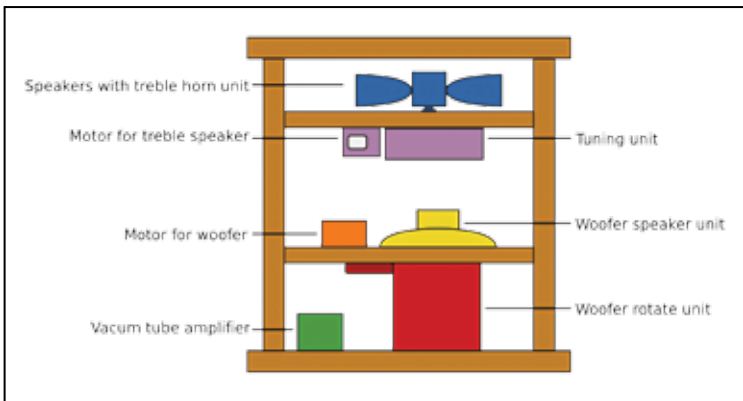


pitch matter more than sound quality, and where space constraints prevent the usage of more traditional speaker technologies. For instance, applications in sonar in which piezoelectric devices are used as both output devices and input devices are in wide use.

Magnetostatic loudspeakers, on the other hand, use large currents instead of large voltages to produce sound. The arrangement of the permanent magnets in a magnetostatic speaker is similar to an electrostatic speaker; the difference is in the construction of the diaphragm. Wires and strips carrying current are bonded to the diaphragm, and these strips cause the diaphragm to move along them in the presence of the magnetic field. Because of the absence of any obstruction behind and in front of the diaphragm, these speakers are capable of breathing freely to the front and rear, this being able to provide better sound quality. However, these class of speakers are quite large in size and width, so there exists only a niche market for such speakers.

Then there are plasma arc speakers, most popular for not using a diaphragm to begin with. Plasma speakers work on the same principle of creating a variation in air pressure to produce sound. What sets them apart is the use of high-energy electrical plasma that is used to vary air pressure to produce sound, instead of a solid diaphragm. In conventional speakers, the mass of the diaphragm is what causes the main problem. The inertia inherent to its mass resists acceleration, thus causing tensile fatigue at one point. In comparison, the low-inertia driver of a plasma arc speaker (the plasma itself) has an exceptionally quick response compared



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